Category

Yield and Yield Spread Measures for Floating-Rate Instruments

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YIELD MEASURES FOR MONEY MARKET INSTRUMENTS

Learning Outcome

- calculate and interpret yield measures for money market instruments

Money market instruments are debt securities with original maturities of one year or less. There are many types, including overnight sale and repurchase agreements (repos), bank certificates of deposit, commercial paper, Treasury bills (government securities issued with a maturity of one year or less), bankers' acceptances, and time deposits based on market reference rates. There are mutual funds that invest solely in eligible money market securities, known as money market mutual funds, which are sometimes considered as an alternative to bank deposits.

There are several important differences in yield measures quoted for money market instruments versus bonds:

1. Bond yields-to-maturity are annualized and compounded. Yield measures in the money market are annualized but *not* compounded; the return on a money market instrument is stated on a simple interest basis.
2. Bond yields-to-maturity usually are stated for a common periodicity for all times-to-maturity. Money market instruments with different times-to-maturity have different periodicities for the annual rate.
3. Bond yields-to-maturity can be calculated using standard time-value-of-money analysis. Money market instruments are often quoted using non-standard interest rates and require different pricing equations than those used for bonds.

In general, quoted money market rates are either **discount rates** or **add-on rates**. Although market conventions vary around the world, commercial paper, Treasury bills, and bankers' acceptances often are quoted on a discount rate basis. Bank certificates of deposit, repos, and market reference rate indexes are quoted on an add-on rate basis. While *discount rate* usually means "interest rate used to calculate a present value," in the money market, the discount rate involves an instrument for which interest is included in the face value of the instrument. An add-on rate, in contrast, involves interest that is added to the principal or investment amount. Some examples will clarify this point.

Equation 2 is the pricing formula for money market instruments quoted on a discount rate basis.

$$PV = FV \times \left(1 - \frac{\text{Days}}{\text{Year}} \times DR\right), \qquad 2$$

where

PV = present value, or the price of the money market instrument

FV = the future value paid at maturity, or the face value of the money market instrument

Days = the number of days between settlement and maturity

Year = the number of days in the year

DR = the discount rate, stated as an annual percentage rate

Suppose that a 91-day Indian Treasury bill (T-bill) with a face value of INR10 million is quoted at a discount rate of 3.45% for an assumed 360-day year. The price of the Indian rupee T-bill is INR9,912,792.

$$PV = 9,912,791.67.$$

The unique characteristics of a money market discount rate can be examined with Equation 3, which transforms Equation 2 algebraically to isolate the *DR* term.

$$DR = \frac{\text{Year}}{\text{Days}} \times \frac{(FV - PV)}{FV}. \qquad 3$$

The first term, Year/Days, is the periodicity of the annual rate. The second term reveals the odd character of a money market discount rate. The numerator, *FV* – *PV*, is the interest earned on the rupee T-bill, INR87,208 (= 10,000,000 – 9,912,792), over the 91 days to maturity. However, the denominator is *FV*, not *PV*. In theory, an interest rate is the amount earned divided by the investment amount (*PV*), not the maturity value (*FV*), which includes the earnings. Therefore, by design, a money market discount rate *understates* the rate of return to the investor, and it *understates* the cost of borrowed funds to the issuer. That is because *PV* is less than *FV* if *DR* is greater than zero.

Equation 4 is the pricing formula for money market instruments quoted on an add-on rate basis.

$$PV = \frac{FV}{(1 + \frac{\text{Days}}{\text{Year}} \times AOR)}, \qquad 4$$

where

PV = present value, the principal amount, or the price of the money market instrument

FV = the future value, or the redemption amount paid at maturity including interest

Days = the number of days between settlement and maturity

Year = the number of days in the year

AOR = the add-on rate, stated as an annual percentage rate

EXAMPLE 2

Yield Measures on Bank Certificates of Deposit

Commercial Finance Partners AG (CFP Bank) issues 90-day certificates of deposit (CDs) to fund its floating-rate loans made to customers. The objective is to maintain a match between the bank's assets (floating-rate loans) and its funding sources. The quoted add-on rate for its 90-day CD is 0.12%, assuming a 365-day year. If the initial principal amount is EUR20 million, the redemption amount due at maturity is found by rearranging Equation 4 and entering *PV* = 20,000,000, Days = 90, Year = 365, and *AOR* = 0.0012.

$$FV = 20,000,000 + (20,000,000 \times \frac{90}{365} \times 0.0012).$$

$$FV = 20,005,918.$$

At maturity, CFP Bank pays EUR20,005,918, the principal of EUR20,000,000 plus interest of EUR5,918. The interest is calculated as the principal times the fraction of the year times the annual add-on rate. It is added to the principal to determine the redemption amount.

Suppose that after 45 days, the CD investor sells the CD to a dealer. At that time, the quoted add-on rate for a 45-day CD is 0.12%. The sale price for the CD can be calculated using Equation 4 for *FV* = 20,005,918, Days = 45, Year = 365, and *AOR* = 0.0012. The sale price is EUR20,004,438.

$$(1 + \frac{45}{365} \times 0.0012)$$

$$PV = 20,004$$
, 438.

The characteristics of an add-on rate can be examined with Equation 5, which transforms Equation 4 algebraically to isolate the AOR term.

$$AOR = \frac{\text{Year}}{\text{Days}} \times \frac{FV - PV}{PV}.$$

The first term, Year/Days, is the periodicity of the annual rate. The second term is the interest earned (*FV* – *PV*) divided by the amount invested (*PV*).

CFP Bank's rate of return on its 45-day investment in the certificates of deposit can be calculated with Equation 5.

$$AOR = \frac{\text{Year}}{\text{Days}} \times \frac{FV - PV}{PV}.$$

$$AOR = \frac{365}{45} \times \frac{20,005,918 - 20,004,438}{20,004,438}.$$

$$AOR = 0.0006.$$

The rate of return, stated on a 365-day add-on rate basis, is 0.06%. This result is an annual rate for a periodicity of 8.11 (= 365/45). Implicitly, this assumes that the investment can be replicated 8.11 times over the year.

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Investment analysis is more challenging for money market securities because (1) some instruments are quoted on a discount rate basis while others on an add-on rate basis and (2) some assume a 360-day year and others use a 365-day year. Another difference is that the "amount" of a money market instrument quoted by traders on a discount rate basis typically is the face value paid at maturity. However, the "amount" when quoted on an add-on rate basis usually is the price at issuance. Exhibit 2 summarizes these conventions.

Exhibit 2:

Summary of Money Market Yield Measures

Conventions and Instruments	Description	
Quote Basis	Discount Rate (DR)	Add-on Rate (AOR)
Equation	*DR* = (Year/Days) × [(*FV* – *PV*)/*FV*].	*AOR* = (Year/Days) × [(*FV* – *PV*)/*PV*].
Quoted Amount	Face Value at Maturity (FV)	Price at Issuance (PV)
Typical Instruments	Commercial Paper, Treasury Bills, Bankers' Acceptances	Bank Certificates of Deposit, Repurchase Agreements (Repos), Market Reference Rates (MRRs)

To make money market investment decisions, it is essential to compare instruments on a common basis. The following example illustrates this point.

EXAMPLE 3

Yield Measures on Bank Certificates of Deposit

Suppose that an investor is comparing the following two money market instruments. Which offers the higher expected rate of return, assuming that credit risks are the same?

0.120% for a 365-day year

The price of the commercial paper is 98.560 per 100 of face value, calculated using Equation 2 and entering *FV* = 100, Days = 90, Year = 360, and *DR* = 0.0012.

$$PV = FV \times (1 - \frac{\text{Days}}{\text{Year}} \times DR).$$

$$PV = 100 \times (1 - \frac{90}{360} \times 0.0012).$$

$$PV = 99.970.$$

Next, use Equation 5 to solve for *AOR* for a 365-day year, where Year = 365, Days = 90, *FV* = 100, and *PV* = 99.970.

$$AOR = \frac{\text{Year}}{\text{Days}} \times \frac{FV - PV}{PV}.$$

$$AOR = \frac{365}{90} \times \frac{100 - 99.970}{99.970}.$$

$$AOR = 0.00122.$$

The 90-day commercial paper discount rate of 0.120% converts to an add-on rate for a 365-day year of 0.122%. This converted rate is called a bond equivalent yield, or sometimes just an "investment yield." A **bond equivalent yield** is a money market rate stated on a 365-day add-on rate basis. If the risks are the same, BRWA's CP offers 0.2 bps more in annual return than CFP Bank's CD.

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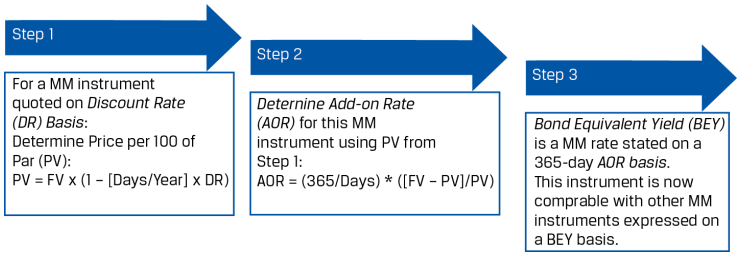
Exhibit 3 summarizes the process just illustrated for comparing money market instruments on a common, bond equivalent yield basis.

Exhibit 3:

Comparing Money Market Instruments on Bond Equivalent Yield Basis



▶ Expand / collapse Extended Description

EXAMPLE 4

Comparing Money Market Instruments Based on Bond Equivalent Yields

A money market investor observes quoted rates on four 180-day money market instruments: A, B, C, and D.

Money Market Instrument	Quotation Basis	Assumed Number of Days in the Year	Quoted Rate
A	Discount Rate	360	4.33%
B	Discount Rate	365	4.36%
C	Add-On Rate	360	4.35%

Using a spreadsheet, we can calculate the bond equivalent yield for each instrument so that rates of return can be compared. Bond A offers the highest rate of return (4.487%) and Bond C offers the lowest rate of return (4.410%) on a bond equivalent yield basis. Please refer to the candidate learning ecosystem online for a spreadsheet demonstrating the calculation in the downloadable Excel workbook.

	Money market Instrument	A	B	C	D
Input	Quotation Basis	Discount Rate	Discount Rate	Add-On Rate	Add-On Rate
	Number of Days in the Year	360	365	360	365
	Quoted Rated (%)	4.33%	4.36%	4.35%	4.45%
Output	PV/FV	97.835	97.850	102.175	102.195
	Add-On-Rate	0.04487	0.04456	0.04410	0.04450
	Bond Equivalent Yield (%)	4.487%	4.456%	4.410%	4.450%

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Another important difference between yield measures in the money market and the bond market is the periodicity of the annual rate. Because bond yields-to-maturity are computed using interest rate compounding, there is a well-defined periodicity. For instance, bond yields-to-maturity for semiannual compounding are annualized for a periodicity of two. Money market rates are computed using simple interest without compounding. In the money market, the periodicity is the number of days in the year divided by the number of days to maturity. Therefore, money market rates for different times-to-maturity have different periodicities.

Suppose that an analyst prefers to convert money market rates to a semiannual bond basis so that the rates are directly comparable to yields on bonds that make semiannual coupon payments. The quoted rate for a 91-day Indian rupee T-bill is 3.50%, quoted as a bond equivalent yield, which means its periodicity is 365/91. We use the periodicity conversion formula covered in a previous lesson to convert from a periodicity *m* of 365/91 to *n* of 2:

$$\left(1+\frac{APR_m}{m}\right)^m = \left(1+\frac{APR_n}{n}\right)^n.$$

$$\left(1+\frac{0.035}{\frac{365}{91}}\right)^{\frac{365}{91}} = \left(1+\frac{APR_2}{2}\right)^2.$$

$$APR_2 = 0.03515 = 3.515\%.$$

Therefore, 3.50% for a periodicity of 365/91 corresponds to 3.515% for a periodicity of 2. The difference is –1.5 bps. In general, the difference depends on the level of the annual percentage rate. When interest rates are lower, the difference between the annual rates for any two periodicities is reduced.

Question Set

1. Identify the correct relationship for a money market instrument quoted on a discount rate basis:

A. $PV = FV \times (1 - \frac{\text{Days}}{\text{Year}} \times DR)$.

B. $FV = PV \times (1 - \frac{\text{Year}}{\text{Days}} \times DR)$.

Solution:

A is correct. In the money market, the discount rate is a specific type of quoted rate. Hence, the price or the present value of a money market instrument is determined by

$$PV = FV \times (1 - \frac{\text{Days}}{\text{Year}} \times DR),$$

where

PV = present value, or the price of the money market instrument

FV = the future value paid at maturity, or the face value of the money market instrument

Days = the number of days between settlement and maturity

Year = the number of days in the year

DR = the discount rate, stated as an annual percentage rate

2. *True or false:* Yield measures in money market instruments are annualized and compounded.

A. True
B. False

Hide Solution

Solution:

False. Bond yields-to-maturity are annualized and compounded. However, yield measures in money markets are annualized but not compounded.

3. *True or false:* Money market instruments use the standard time value of money, such as discount rates or add-on rates.

A. True
B. False

Hide Solution

Solution:

False. Bond yields-to-maturity are calculated using the standard time value of money. Money market instruments are often quoted using non-standard interest rates, such as discount rates or add-on rates.

4. *True or false:* Money market instruments with different times-to-maturity have different periodicities.

A. True
B. False

Hide Solution

Solution:

True. Money market instruments with different times-to-maturities have different periodicities. However, bond yields-to-maturity are given for a common periodicity for all times-to-maturity.

5. Explain the concept of a bond equivalent yield for a money market instrument.

Hide Solution

Solution:

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earned on the money market instrument with a similar return earned by a bond.

6. A portfolio manager has asked you to evaluate the following Thai baht–denominated 180 days money market instruments with equivalent credit risk.

Money Market Instrument	Quotation Basis	Number of Days in the Year	Quoted Rate
Short-Term Thai T-Bills	Discount Rate	360	4.78%
Thai Bank Certificate of Deposit	Discount Rate	365	4.81%
Thai Corporate Commercial Paper	Add-On Rate	365	4.85%

Which instrument has the highest rate of return on a bond equivalent yield basis?

A. Short-term Thai T-bills
B. Thai bank certificate of deposit
C. Thai corporate commercial paper

Hide Solution

Solution:

A is correct. The bond equivalent yield for each instrument is most easily calculated using a spreadsheet. Please refer to the candidate learning ecosystem online for a spreadsheet demonstrating the calculations in the downloadable Excel workbook. The short-term Thai T-bills offer the highest return of 4.965% on a bond equivalent yield basis.

	Money Market Instrument	Short-Term Thai T-Bill	Thai Bank CD	Thai Corporate CP
Input	Quotation Basis	Discount Rate	Discount Rate	Add-On Rate
	Number of Days in the Year	360	365	365
	Quoted Rated (%)	4.78%	4.81%	4.85%
Output	PV/FV	97.610	97.628	102.392
	Add-On-Rate	0.04965	0.04927	0.04850
	Bond Equivalent Yield (%)	4.965%	4.927%	4.850%

The following information relates to questions 7-8

A German insurance company buys a 91-day certificate of deposit with a quoted add-on rate of 0.40% for a 365-day year.

7. If the initial payment amount is EUR20 million, calculate the redemption amount due at maturity.

Hide Solution

Solution:

The redemption amount can be calculated using Equation 4:

$$PV = \frac{FV}{(1 + \frac{\text{Days}}{\text{Year}} \times AOR)}.$$

$$20,000,000 = \frac{FV}{(1 + \frac{91}{365} \times 0.0040)}.$$

$$FV = 20,019,945.$$

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Hide Solution

Solution:

After 31 days, the sale price of the certificate of deposit is also calculated using Equation 4:

$$PV = \frac{FV}{(1 + \frac{\text{Days}}{\text{Year}} \times AOR)}.$$

$$PV = \frac{20,019,945}{(1 + \frac{60}{365} \times 0.0021)}.$$

$$PV = 20,013,036.$$

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OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

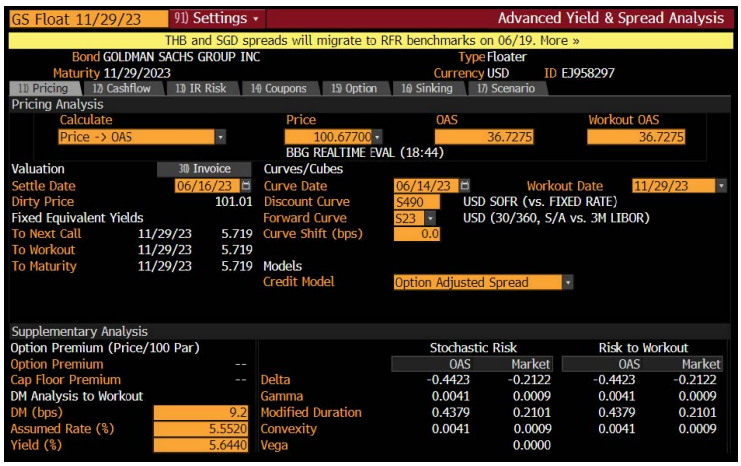
Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{YASN} ADVANCED YIELD & SPREAD ANALYSIS

The Bloomberg YASN function allows you to view yield and spread measures for floating-rate notes that include embedded options.

> "YASN <GO> provides a standardized yield analysis calculator for structured notes, so you can price and analyze the risk of a bond with complex embedded options and callability features. YASN supports the Hull-White One-Factor (HW1F) model, which analyzes interest rates to estimate the probability that the options will be exercised on a given call date. For products highly sensitive to interest rate changes, YASN monitors the option premium, allowing you an easy way to evaluate the prices of the options embedded in the structure. When analyzing a bond, you can customize the inputs of importance to you to include such measures as assumptions, yields, fixed-equivalent yields, and risk and hedge ratios." -Bloomberg

To explore this function, type YASN into the Bloomberg command line and press Enter or <GO>. Choose one of the examples offered.



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Please provide feedback on this Bloomberg feature.

What do you want to discuss... (required)

MO

Practice questions, the first question is wrong. For the CFA staff — I am not going to submit errata as it is a very tedious process.

The answer is NOT B, it is C.

A 2-year floating-rate note issued by a French corporation pays the three-month MRR of –0.55% plus 160 bps. The floater is priced at 101.20 per 100 of par value. Assuming the 30/360 day-count convention and 90 days per period, the discount margin for the floater is closest to:

A. 25 bps.

B. 50 bps.

C. 110 bps.

Cash Flow & Payment:

3-months MRR = -0.55% = -0.0055 (annual)

Quoted Margin = 160bps = 0.016 (annual)

So, annual coupon = MRR + Quoted Margin = -0.0055 + 0.016 = 1.05% ANNUAL

Payments are quarterly (m = 4), so periodic PMT (per $100 par) = Annual Coupon / 4 * 100

Periodic PMT = 0.0105 / 4 * 100 = $0.2625

Using financial calculator we solve for the discount rate:

N = 2 * 4 = 8

PV = -101.20

PMT = 0.2625

FV = 100

CPT I/Y = 0.111745% PER QUARTER, i.e., 0.00111745 in decimals

Calculating Discount Margin:

r = (MRR + DM) / m

0.00111745 = (-0.0055 + DM) / 4

DM = 0.0099698 OR 99.698bps, which is closest to C. 110 bps.

Created 3 months ago by Maxim Ovchinnikov 0 replies | Last Activity: 3 months ago

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MO

Same thing for Q7 and Q8, using LOGIC and not having to memorize formulas.

Q7. If the initial payment amount is EUR20 million, calculate the redemption amount due at maturity.

Convert the annual 0.4% to 91-day period rate = 91/365 * 0.004 = 0.00099726

FV = 20mil * (1 + 0.00099726) = 20,019,945.20

Q8. Suppose that after 31 days, the insurance company sells the CD. At the time of the sale, the quoted add-on rate for a 60-day CD is 0.21%. Calculate the sale price for the CD.

Calculate the add on factor = 0.0021 * 60/365 = 0.00034521

Here, we're calculating the Sale Price, i.e., PV = 20,019,945.20 / (1 + 0.00034521) = 20,013,036.49

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MO

Logical solution to Q6 without having to memorize all the formulas.

Short-term Thai T-bills:

180/360 * 0.0478 = 0.0239

PV = 100 * (1 – 0.0239) = 97.61

So, you pay 97.61 now and get 100 at maturity => imbedded interest = 100 – 97.61 = 2.39

180-day investor yield = 2.39 / 97.61 = 0.0244852

Bond Equivalent Yield (%) = Annualized (365) = 0.0244852 * (365/180) = 4.96505367%

Thai bank certificate of deposit:

180/365 * 0.0481 = 0.02372055

PV = 100 * (1 - 0.02372055) = 97.62794521

So, you pay 97.62794521 now and get 100 at maturity => imbedded interest = 100 – 97.62794521 = 2.37205479

180-day investor yield = 2.37205479 / 97.62794521 = 0.02429688

Bond Equivalent Yield (%) = Annualized (365) = 0.02429688 * (365/180) = 4.92686801%

So, you pay 100 now and get 102.391781 at maturity => add-on interest = 102.391781 – 100 = 2.391781
180-day investor yield = 2.391781 / 100 = 0.02391781
Bond Equivalent Yield (%) = Annualized (365) = 0.02391781 * (365/180) = 4.85%

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PZ

CAN ANYONE EXPLAIN HOW Q8 OF THE PRACTICE QUESTIONS THE RESULT IS 0.0028? it should be 0.0056. i dont understand why they divided it by 2. and even if they did, why are they not showing the division

Screenshot%202025-09-13%20at%2013.26.14.png (217 KB)

Created 4 months ago by PIERFRANCESCO ZEOLI — 2 replies | Last Activity: 3 months ago
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AZ

anyone wondering why the examples are mostly wrong - check this link (https://www.cfainstitute.org/sites/default/files/-/media/documents/support/programs/cfa/2025-cfa-level-i-errata.pdf) it contains all the edits for this lesson and all other lessons. hope this help!

Created 4 months ago by Abdulaziz Zahid — 0 replies | Last Activity: 4 months ago
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KB

Question Set - Q8
Can someone please help me understand this question? what is going on in this question?

Created 4 months ago by Krisha Bhavin Jain — 0 replies | Last Activity: 4 months ago
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JJ

Whatever is going on at example 3 needs to be fixed.

Created 7 months ago by Jacobus Johannes Petrus Du Plessis — 1 reply | Last Activity: 4 months ago
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AA

Exampe3 :The quoted price of 98.560 for the commercial paper appears to be incorrect, given the stated 90 day discount rate of 0.10% on a 360 day basis. Using the correct formula and inputs, the price should be approximately 99.975. It seems the price was mistakenly calculated using a much higher discount rate (1.2%) instead of 0.10%. Can someone confirm or refute this?

Created 8 months ago by Abdellah Asmahri — 6 replies | Last Activity: 4 months ago
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AP

"Treasury bills, and bankers' acceptances often are quoted on a discount rate basis. Bank certificates of deposit, repos, and market reference rate indexes are quoted on an add-on rate basis"? I think it is opposite am i right?

Created 5 months ago by Anujin Purevjav — 0 replies | Last Activity: 5 months ago
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DN

The Question no 6, don't we have to annualize the AOR?

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